

```
1  import java.util.Scanner;
2  public class Numbersfrom1toN {
3      public static void main(String[] args){
4          Scanner sc = new Scanner(System.in);
5          System.out.println("Enter a number: ");
6          int n = sc.nextInt();
7
8          for(int i = 1; i <= n; i++){
9              System.out.print(i+ " ");
10
11          }
12      }
13  }
```

Run

Numbersfrom1toN x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\j

Enter a number:

15

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Process finished with exit code 0

```
1 import java.util.Scanner;
2 public class Multipleofthree {
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("Enter a number: ");
6         int n = sc.nextInt();
7         for(int i = 3; i <= n; i = i + 3){
8             System.out.println(i+ " ");
9         }
10    }
11 }
12
```

Run

Multipleofthree x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\jav

Enter a number:

40

3

6

9

12

15

18

21

24

27

30

33

36

39

Process finished with exit code 0

 MenuDrivenProgram.java reversetriangle.java Numbersfrom1toN.java Multipleofthree.java

```
1  import java.util.Scanner;
2  ▶ public class Factorial {
3  ▶      public static void main(String[] args){
4          Scanner sc = new Scanner (System.in);
5          System.out.print("Enter a number: ");
6          int n = sc.nextInt();
7          int factorial = 1;
8          for(int i = 1; i <= n; i++) {
9              factorial = factorial * i;
10         }
11         System.out.println("Factorial of" + n + " is " + factorial);
12
13     }
14 }
15
```

C:\Users\Sakshi\.jdk\openjdk-26.0.

Enter a number: 9

Factorial of 9 is 362880

Process finished with exit code 0

```
1 import java.util.Scanner;
2 public class Evennumbers {
3     public static void main(String[] args){
4         Scanner sc = new Scanner (System.in);
5         System.out.println("Enter a number: ");
6         int n = sc.nextInt();
7
8         for (int i = 2; i <= n; i = i + 2){
9
10            System.out.println(i+ " ");
11        }
12    }
13
14 }
```



C:\Users\Sakshi\.jdk\ope

Enter a number:

20

2

4

6

8

10

12

14

16

18

20

ava

Numbersfrom1toN.java

Multipleofthree.java

Factorial.java

```
import java.util.Scanner;
public class Sumofoddnum {
    public static void main(String[] args){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;
        for(int i = 1; i <= n; i = i + 2){
            sum = sum + i;
        }
        System.out.println(" sum of odd number " + n + " is " +sum)
    }
}
```

Run

Sumofoddnum x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\

Enter a number:

12

sum of odd number 12 is 36

Process finished with exit code 0

```
1  import java.util.Scanner;
2
3  public class Printanarray {
4      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          int[] numbers = new int[5];
9
10         System.out.println("Enter 5 integers:");
11
12         for (int i = 0; i < 5; i++) {
13             numbers[i] = sc.nextInt();
14         }
15
16         for (int number : numbers) {
17             System.out.print(number + " ");
18         }
19     }
20 }
```

Run Printanarray x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin
Enter 5 integers:
8 14 15 18 20
8 14 15 18 20
Process finished with exit code 0

```
1  import java.util.Scanner;
2  ▶ public class sumofelementsinarrray {
3  ▶     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[] numbers = new int[5];
6         System.out.println("Enter 5 integers:");
7         for (int i = 0; i < 5; i++) {
8             numbers[i] = sc.nextInt();
9         }
10        int sum = 0;
11        for (int number : numbers) {
12            sum = sum + number;
13        }
14        System.out.println("The sum of all numbers is: " + sum);
15    }
16 }
```

Run

sumofelementsinarray x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\

Enter 5 integers:

2 4 6 8 10

The sum of all numbers is: 30

Process finished with exit code 0

```
1 import java.util.Scanner;
2 public class PrintNames {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String[] names = new String[5];
6         System.out.println("Enter 4 names:");
7         for (int i = 0; i < 5; i++) {
8             names[i] = sc.next();
9         }
10        for (String name : names) {
11            System.out.println(name);
12        }
13    }
14 }
```



C:\Users\Sakshi\.jdk\openjdk-26.0.

Enter 4 names:

Sakshi Pranav Sumit Shruti Om

Sakshi

Pranav

Sumit

Shruti

Om


```
1 import java.util.Scanner;
2 import java.util.Arrays;
3 public class LargestElement {
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6         int[] numbers = new int [5];
7         System.out.println("Enter 5 integer: ");
8         for(int i = 0; i < 5; i++){
9             numbers[i] = sc.nextInt();
10        }
11        Arrays.sort(numbers);
12        System.out.println(" Largest element is " + numbers[4]);
13    }
14 }
```

Run

LargestElement x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java

Enter 5 integer:

10 20 30 58 47

Largest element is 58

Process finished with exit code 0

|

```
1 import java.util.Scanner;
2 public class Averageofnumbers {
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int[] numbers = new int [5];
6         System.out.println("Enter 5 integer: ");
7         for(int i = 0; i < 5; i++){
8             numbers[i] = sc.nextInt();
9         }
10        int sum = 0;
11        for(int number : numbers){
12            sum = sum + number;
13        }
14        double average = (double) sum/5;
15        System.out.println("Average of number: " + average);
16    }
17 }
```

Run

Averageofnumbers x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\j

Enter 5 integer:

20 30 60 80 45

Average of number: 47.0

Process finished with exit code 0

|

```
1 import java.util.Scanner;
2 public class positiveornegative {
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int[] numbers = new int[6];
6         System.out.println("Enter 6 numbers: ");
7         for(int i = 0; i < 6; i++){
8             numbers[i] = sc.nextInt();
9         }
10        int positive = 0;
11        int negative = 0;
12        for(int number : numbers){
13            if(number > 0){
14                positive++;
15            }
16            else if (number < 0){
17                negative++;
18            }
19        }
20        System.out.println("Positive number: " + positive);
21        System.out.println("Negative number: " + negative);
22    }
23 }
24
```

Run

positiveornegative x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin

Enter 6 numbers:

-6 85 -90 0 -63 25

Positive number: 2

Negative number: 3

Process finished with exit code 0

|

 PrintNames.java LargestElement.java Averageofnumbers.java

```
1  import java.util.Scanner;
2  import java.util.Arrays;
3  ▶ public class ascendingorder {
4  ▶     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6         int[] numbers = new int[5];
7         System.out.println("Enter 5 number: ");
8         for(int i = 0; i < 5; i++){
9             numbers[i] = sc.nextInt();
10        }
11        Arrays.sort(numbers);
12        System.out.print("Sorted array: ");
13        for(int number : numbers){
14            System.out.print(number + " ");
15        }
16    }
17 }
```

Run ascendingorder x





```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\
Enter 5 number:
45 85 20 13 84
Sorted array: 13 20 45 84 85
Process finished with exit code 0
|
```


LargestElement.java

Averageofnumbers.java

positiveornegative.java

```
1 import java.util.Scanner;
2 import java.util.Arrays;
3 public class searchnumber {
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6         Integer[] numbers = new Integer[5];
7         System.out.println("Enter 5 inetegers: ");
8         for(int i = 0; i < 5; i++){
9             numbers[i] = sc.nextInt();
10        }
11        System.out.println("Enter the number to search: ");
12        int search = sc.nextInt();
13
14        if (Arrays.asList(numbers).contains(search)){
15            System.out.println("Found");
16        }else {
17            System.out.println("Not found ");
18        }
19    }
20 }
```

Run

searchnumber x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java

Enter 5 inetegers:

20 25 85 60 90

Enter the number to search:

85

Found

Process finished with exit code 0

```
1 import java.util.Scanner;
2 import java.util.Arrays;
3 public class findindex {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] numbers = new int[5];
7         System.out.println("Enter 5 integers:");
8         for (int i = 0; i < 5; i++) {
9             numbers[i] = sc.nextInt();
10        }
11        System.out.print("Enter the number to search: ");
12        int search = sc.nextInt();
13        int index = Arrays.binarySearch(numbers, search);
14        if (index >= 0) {
15            System.out.println("The number " + search + " is found at index " + index);
16        } else {
17            System.out.println("Not Found");
18        }
19    }
20 }
```

Run

findindex x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\jav

Enter 5 integers:

58 42 10 17 85

Enter the number to search: 10

The number 10 is found at index 2

Process finished with exit code 0

|

```
1  ▶ public class pattern1 {
2  ▶     public static void main(String[] args) {
3      for (int i = 1; i <= 5; i++) {
4          for (int j = 1; j <= i; j++) {
5              System.out.print(i);
6              if (j < i) {
7                  System.out.print("*");
8              }
9          }
10         System.out.println();
11     }
12     for (int i = 5; i >= 2; i--) {
13         for (int j = 1; j <= i; j++) {
14             System.out.print(i);
15             if (j < i) {
16                 System.out.print("*");
17             }
18         }
19         System.out.println();
20     }
21 }
22 }
```

Run

pattern1 x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin

1

2*2

3*3*3

4*4*4*4

5*5*5*5*5

5*5*5*5*5

4*4*4*4

3*3*3

2*2

Process finished with exit code 0

positiveornegative.java

ascendingorder.java

searchnumber.java

```
1 public class pattern2 {  
2     public static void main(String[] args) {  
3         for (int i = 1; i <= 5; i++) {  
4             for (int j = 1; j <= i; j++) {  
5                 System.out.print(j);  
6                 if (j < i) {  
7                     System.out.print("*");  
8                 }  
9                 }  
10                System.out.println();  
11            }  
12        }  
13    }  
14 }
```

C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\jav

1

1*2

1*2*3

1*2*3*4

1*2*3*4*5

Process finished with exit code 0


```
1  ▶ public class pattern3 {
2  ▶     public static void main(String[] args) {
3
4         for (int i = 1; i <= 5; i++) {
5
6             for (int j = 1; j <= i; j++) {
7                 System.out.print((2 * j - 1));
8
9                 if (j < i) {
10                     System.out.print("*");
11                 }
12             }
13
14             System.out.println();
15         }
16     }
17 }
```

Run

pattern3 x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin

1

1*3

1*3*5

1*3*5*7

1*3*5*7*9

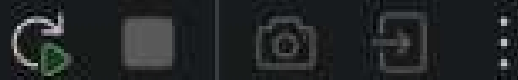
Process finished with exit code 0

```
public class pattern4 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i++) {  
  
            for (int j = 1; j <= 5; j++) {  
                System.out.print(i);  
            }  
            System.out.println();  
        }  
    }  
}
```



Run

pattern4 x



```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\
```

```
11111
```

```
22222
```

```
33333
```

```
44444
```

```
55555
```

```
Process finished with exit code 0
```

index.java

mainindex.java

pattern1.java

pattern2.java

```
1  ▶ public class pattern5 {  
2  ▶     public static void main(String[] args) {  
3  
4         for (int i = 1; i <= 5; i++) {  
5  
6             for (int j = 1; j <= i; j++) {  
7                 System.out.print(i);  
8             }  
9             System.out.println();  
10        }  
11    }  
12 }
```

Run pattern5 x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin
1
22
333
4444
55555

Process finished with exit code 0

index.java

pattern1.java

pattern2.java

pattern3.java

```
1  ▶ public class pattern6 {  
2  ▶     public static void main(String[] args) {  
3        
4          for (int i = 1; i <= 5; i++) {  
5                
6                  for (int j = 1; j <= i; j++) {  
7                  System.out.print(j);  
8                  }  
9                  System.out.println();  
10             }  
11         }  
12     }
```



C:\Users\Sakshi\.jdk\openjdk-26.0.2\b

1

12

123

1234

12345

Process finished with exit code 0

index.java

pattern1.java

pattern2.java

pattern3.java

```
1  ▶ public class pattern6 {  
2  ▶     public static void main(String[] args) {  
3      int number = 1;  
4      for (int i = 1; i <= 5; i++) {  
5            
6          for (int j = 1; j <= i; j++) {  
7              System.out.print(number + " ");  
8              number++;  
9          }  
10         System.out.println();  
11     }  
12 }  
13 }
```



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

Process finished with exit code 0

```
1  ▶ public class pattern7 {  
2  ▶     public static void main(String[] args){  
3      int n = 6;  
4      for(int i = 1; i <= n; i++){  
5          for(int j = 1; j <= n; j++){  
6              if(i == 1 || i == n || j == 1 || j == n){  
7                  System.out.print("*");  
8              }else{  
9                  System.out.print(" ");  
10             }  
11         }  
12     System.out.println();  
13     }  
14 }  
15 }  
16
```

Run

pattern7 x



```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin
```

```
*****
```

```
*      *
```

```
*      *
```

```
*      *
```

```
*      *
```

```
*****
```

```
Process finished with exit code 0
```

```
1  ▶ public class pattern8 {
2  ▶     public static void main(String[] args){
3      int n = 6;
4      for (int i = 1; i <= n; i++){
5          for(int j = 1; j <= n - i; j++){
6              System.out.print(" ");
7          }
8          for(int j = 1; j <= (2 * i - 1); j++){
9              System.out.print("*");
10         }
11         System.out.println();
12     }
13     for(int i = n - 1; i >= 1; i--){
14         for(int j = 1; j <= n - i; j++){
15             System.out.print(" ");
16         }
17         for(int j = 1; j <= (2 * i - 1); j++){
18             System.out.print("*");
19         }
20         System.out.println();
21     }
22 }
23 }
```



C:\Users\Sakshi\.jdk\openjdk-26.0.2\

*

*

Process finished with exit code 0

3.java pattern4.java pattern5.java pattern6.java pattern7.java

```
1  import java.util.Scanner;
2  public class reversestring {
3      public static void main(String[] args){
4          Scanner sc = new Scanner(System.in);
5          System.out.println("Enter s string: ");
6          String str = sc.nextLine();
7          String reverse = "";
8          for(int i = str.length()- 1; i >= 0; i--){
9              reverse = reverse + str.charAt(i);
10         }
11         System.out.println("Reverse string: " + reverse);
12     }
13 }
14
```

Run

reversestring x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\b

Enter s string:

reverse

Reverse string: esrever

Process finished with exit code 0

|

pattern4.java pattern5.java pattern6.java pattern7.java reversestring.java

```
1 import java.util.Scanner;
2 public class countvowels {
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("Enter a string: ");
6         String str = sc.nextLine();
7         int count = 0;
8         for(int i = 0; i < str.length(); i++){
9             char ch = str.charAt(i);
10            if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u'){
11                count++;
12            }
13        }
14        System.out.println(" the number of vowels is " + str + " is " +count);
15    }
16 }
```

Run

countvowels x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java

Enter a string:

coding

the number of vowels is coding is 2

Process finished with exit code 0

 pattern5.java pattern6.java pattern7.java reversestring.java countvowels

```
1 import java.util.Scanner;
2 public class palindrome {
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("Enter a string: ");
6         String str = sc.nextLine();
7         str = str.replaceAll(regex: "[^a-zA-Z0-9]", replacement: "").toLowerCase();
8         String reverse = "";
9         for(int i = str.length() - 1; i >= 0; i--){
10             reverse = reverse + str.charAt(i);
11         }
12         if(str.equals(reverse)){
13             System.out.println("the string is palindrome");
14         }else{
15             System.out.println("the string is not palindrome");
16         }
17     }
18 }
```

Run

palindrome x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\j

Enter a string:

kanak

the string is palindrome

Process finished with exit code 0

|

pattern6.java

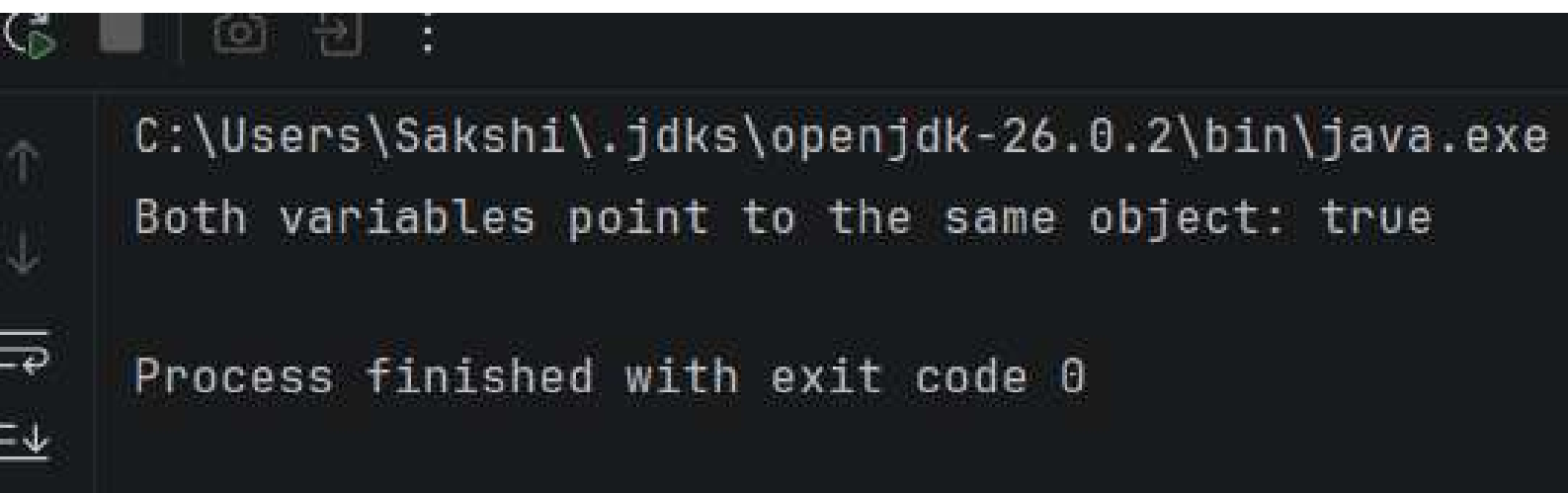
pattern7.java

reversestring.java

countvowels.java

palindrome.java

```
public class stringobject {  
    public static void main(String[] args){  
        String str1 = "bye";  
        String str2 = "bye";  
        System.out.println("Both variables point to the same object: " + (str1 == str2));  
    }  
}
```

A screenshot of a terminal window with a dark background. The window title bar at the top shows standard Windows icons (start, search, task view, volume, network, battery) and a portion of the address bar. The terminal content is as follows:

```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe
Both variables point to the same object: true

Process finished with exit code 0
```

On the left side of the terminal, there is a vertical toolbar with icons for navigating through the command history (up, down, search, etc.).

C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe

Both variables point to the same object: true

Process finished with exit code 0

pattern7.java

reversestring.java

countvowels.java

palindrome.java

stringcreation.java

```
1 public class stringcreation {
2     public static void main(String[] args){
3         String str1 = new String( original: "hello");
4         String str2 = new String( original: "hello");
5         System.out.println("Using == " + (str1 == str2));
6         System.out.println("Using .equals()" + str1.equals(str2));
7     }
8 }
```

Run

stringcreation x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\jav

USing == false

Using .equals()true

Process finished with exit code 0

g.java countvowels.java palindrome.java stringobject.java stringcreation.java StringConcat

```
1 public class StringConcatenation {
2     public static void main(String[] args){
3         String str1 = "hello";
4         String str2 = "bye";
5         String str3 = str1 + str2;
6         System.out.println(" IS str3 pointing to the same object as str1? " + (str3 == str1));
7     }
8 }
9
```

Run

StringConcatenation x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe "-jav
IS str3 pointing to the same object as str1? false

Process finished with exit code 0

```
1 ▶ public class StringIntern {
2 ▶  ⚡ public static void main(String[] args) {
3
4     String str1 = new String( original: "hello");
5     String str2 = str1.intern();
6     String str3 = "hello";
7
8     System.out.println("Is str2 and str3 pointing to the same object? " + (str2 == str3));
9 }
10 }
11
12
```

Run

StringIntern x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java
Is str2 and str3 pointing to the same object?

Process finished with exit code 0

```
1 public class MultipleStrings {  
2     public static void main(String[] args){  
3         String str1 = "java";  
4         String str2 = "java";  
5         String str3 = "java";  
6         System.out.println("All string point to the same object : " +(str1 == str2 && str2 == str3));  
7     }  
8 }  
9
```

Run

MultipleStrings x



C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe

All string point to the same object : true

Process finished with exit code 0